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Assessing the interactions between learning enjoyment, motivation, burnout, and grit in EFL students: a mixed-methods approach

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Abstract

This study examines the relationships among Foreign Language Learning Enjoyment (FLLE), L2 grit, academic burnout, and motivated behavior in English as a Foreign Language (EFL) learners, using a mixed-methods approach. A total of 534 Chinese undergraduate EFL learners participated in the quantitative phase, which used a cross-sectional survey, followed by narrative inquiry with 15 participants in the qualitative phase. Structural Equation Modeling (SEM) analysis revealed that FLLE positively influenced motivated behavior while reducing academic burnout. L2 grit mediated the relationships between FLLE, burnout, and motivation, indicating that enjoyment promotes perseverance and resilience. Burnout was found to negatively impact motivation, underscoring the role of emotional well-being in maintaining engagement. The qualitative findings further supported these results, offering insights into learners' experiences of joy, perseverance, and recovery from burnout. Participants described how moments of enjoyment reignited their long-term motivation, while grit helped them persevere through challenges. Burnout was linked to emotional exhaustion, but positive experiences often helped restore motivation. These findings offer a deeper understanding of the emotional and psychological factors influencing language learning success. The study provides important implications for educators, highlighting the need to create positive learning environments, promote resilience, and address burnout to support students' sustained commitment to language learning.

Keywords Foreign Language Learning enjoyment (FLLE), L2 grit, Academic burnout, Motivated behavior, Mixed-methods, EFL learners, Structural equation modeling (SEM), Narrative inquiry

Introduction

Understanding the emotional and psychological factors that influence second language acquisition (SLA) has gained increasing importance in recent years [1–3]. Positive emotional experiences, resilience, and motivation are crucial elements that shape learners' success in language

learning contexts. Motivation and engagement are widely recognized as key drivers of language learning success [4, 5], but relatively few studies have explored how these factors interact with emotional experiences like enjoyment or resilience-related constructs such as grit and academic burnout [3, 6, 7]. Addressing this gap is essential for developing a comprehensive understanding of how emotional and psychological factors collectively influence language learning outcomes, particularly in English as a Foreign Language (EFL) contexts.

Foreign language learning enjoyment (FLLE) refers to the positive emotional experiences that language

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learners encounter during the process of learning [8, 9]. Research suggests that such positive emotions enhance intrinsic motivation and increase engagement with language tasks, leading to greater persistence and effort in language acquisition [1, 10]. According to Fredrickson's broaden-and-build theory, positive emotions like enjoyment broaden learners' cognitive and emotional resources, enabling them to cope with challenges more effectively [11]. However, while FLLE has been shown to positively affect motivation and resilience [9, 12], its long-term impact on reducing negative outcomes like burnout remains underexplored [13, 14].

Grit, defined as perseverance and passion for long-term goals [15], has emerged as a significant predictor of academic success, including in SLA [7, 16]. Learners with high levels of grit maintain effort and overcome obstacles, which is crucial for success in learning, a process that often requires sustained commitment over extended periods [17, 18]. Exploring how grit mediates the relationship between emotional experiences like FLLE and academic outcomes can provide insights into the mechanisms through which positive emotions influence resilience and sustained motivation in language learning [19–21].

Academic burnout, characterized by emotional exhaustion, cynicism, and a sense of inefficacy [22], poses a significant barrier to learning success [23, 24]. Burnout reduces learners' motivation and engagement, leading to slower progress in language acquisition [25, 26]. Understanding how positive emotional experiences like FLLE can protect learners from burnout is essential for developing strategies to foster resilience and maintain sustained motivation [1, 27, 28]. Despite its significance, the interplay between FLLE and burnout, particularly through the lens of grit, has not been thoroughly investigated in the context of EFL learners.

Motivated behavior in SLA refers to the effort, persistence, and active engagement learners display in their language studies [29]. While prior research has emphasized the importance of both internal factors (such as FLLE and grit) and external factors (such as academic pressures) in shaping motivated behavior [30], the interaction between these factors and burnout remains underexplored, particularly in the EFL context [6, 31]. Investigating these interactions can enhance our understanding of how to support learners' sustained engagement and prevent disengagement due to burnout.

Therefore, this study aims to fill these gaps by examining the relationships among FLLE, L2 grit, academic burnout, and motivated behavior in EFL learners. The novelty of this research lies in its holistic approach, integrating these constructs into a comprehensive model to better understand the emotional and psychological dynamics of successful language learning. Previous

studies have not simultaneously considered these variables within a single framework, and this research seeks to bridge that gap. By exploring how FLLE influences motivated behavior and academic burnout directly and indirectly through L2 grit, the study offers new insights into the mechanisms that underpin language learning success.

In addition to its conceptual contribution, this study employs a mixed-methods design, combining structural equation modeling (SEM) for quantitative analysis and narrative inquiry for qualitative insights. This approach provides a more nuanced understanding of learners' emotional and motivational experiences, enriching the quantitative findings with personal accounts of enjoyment, perseverance, and burnout. The research objectives are as follows:

1. To examine how FLLE directly influences motivated behavior and academic burnout in EFL learners.
2. To investigate the mediating role of L2 grit in the relationship between FLLE and both motivated behavior and academic burnout.
3. To gain deeper insights into the emotional and motivational dynamics of language learning through learners' personal experiences of enjoyment, perseverance, and burnout.

By achieving these objectives, the study contributes to the field by offering practical implications for language educators. Understanding these relationships can inform the development of interventions and teaching strategies that enhance positive emotional experiences, foster resilience through grit, and reduce burnout, ultimately supporting sustained motivation and better language learning outcomes.

Literature review

Theoretical background

This study draws on two key theoretical frameworks: Fredrickson's broaden-and-build theory of positive emotions [32] and Self-Determination Theory (SDT) [5, 33]. Together, these frameworks provide a robust foundation for understanding the interplay among FLLE, motivated behavior, academic burnout, and L2 grit in the context of EFL learning.

Broaden-and-build theory [11, 32] highlights the role of positive emotions, such as enjoyment, in expanding individuals' cognitive and behavioral repertoires while building enduring psychological and emotional resources [34]. In language learning, FLLE—characterized as a positive emotional experience—enables learners to approach challenges with greater openness, creativity, and flexibility. Positive emotions enhance cognitive adaptability and resilience, equipping learners to better manage academic

pressures. This perspective aligns with the study's focus on how FLLE influences motivation and mitigates burnout. By fostering emotional and cognitive resources, FLLE is posited to reduce burnout while promoting sustained motivation and engagement.

SDT [5, 33] complements this perspective by examining the role of intrinsic and extrinsic motivation. Intrinsic motivation, driven by personal interest and satisfaction, supports sustained effort and deeper learning compared to extrinsic motivation, which is tied to external incentives or pressures. According to SDT, three core psychological needs—autonomy, competence, and relatedness—are critical for fostering intrinsic motivation. In the context of this study, FLLE is expected to bolster intrinsic motivation by providing enjoyable and fulfilling learning experiences that satisfy these needs. Furthermore, SDT suggests that intrinsic motivation lowers the likelihood of burnout, as learners find satisfaction in the learning process itself rather than being overly dependent on external outcomes.

Grit, as conceptualized by Duckworth [15], emphasizes perseverance and passion for achieving long-term goals. It comprises two key dimensions: perseverance of effort and consistency of interest, both essential for sustained engagement in challenging tasks such as language learning. Within this study, grit is framed as a resilience-related construct that mediates the relationships between FLLE and motivated behavior, as well as between FLLE and burnout. By promoting long-term commitment, grit enables learners to persist through difficulties, aligning with the broaden-and-build theory's focus on resilience.

The integration of these theoretical frameworks provides a comprehensive understanding of the emotional and motivational dynamics in EFL learning. Fredrickson's broaden-and-build theory [32] explains how FLLE generates positive emotional resources that enhance resilience and engagement, while SDT [5, 33] clarifies how FLLE fosters intrinsic motivation and mitigates burnout by satisfying core psychological needs. Grit complements these perspectives by emphasizing the importance of sustained effort in achieving long-term language learning goals. Together, these frameworks form a cohesive theoretical foundation for the study, linking emotional, motivational, and resilience-related constructs in the context of EFL learning.

The role of FLLE in fostering motivation and engagement

FLLE refers to the positive emotional experiences that learners have during the language learning process, significantly influencing their motivation and engagement [8, 9, 35, 36]. FLLE can be categorized into two main types: social enjoyment, which comes from interactions with peers and teachers, and private enjoyment, which stems from personal satisfaction in mastering language

skills [8, 37]. This highlights both the importance of social connections and the personal rewards of progress in language learning.

Fredrickson's broaden-and-build theory of positive emotions offers insight into FLLE. Positive emotions like enjoyment broaden learners' cognitive and behavioral responses, enhancing resilience and creativity [11]. In the context of language learning, this broadening effect fosters cognitive flexibility and persistence, helping learners build resources to tackle future challenges [38]. Teacher behavior plays a crucial role in promoting FLLE. Enthusiastic, supportive teaching has been shown to enhance enjoyment and reduce anxiety [39, 40]. Teachers who create positive, inclusive classrooms can motivate learners and foster positive attitudes toward language learning [29, 41]. Moreover, strong teacher-student relationships can build trust and comfort, further boosting classroom enjoyment [14, 42]. This supportive dynamic encourages students to engage more deeply with learning materials and participate actively in class.

Social interactions also significantly shape FLLE, especially through collaborative learning. Positive peer relationships and group activities, such as discussions and role-plays, enhance social enjoyment by creating a sense of community [43, 44]. Dewaele and Dewaele [45] found that collaboration and peer support not only improve social enjoyment but also boost overall engagement. These connections with peers motivate students to participate in class and seek additional learning opportunities beyond the classroom [13].

In addition to external influences, individual learner traits also affect FLLE. For example, extraversion has been linked to higher levels of enjoyment, as outgoing students are more likely to enjoy group activities and classroom interactions [45, 46]. Additionally, emotional intelligence—the ability to recognize and manage emotions—plays a key role in fostering FLLE. Learners with high emotional intelligence are better equipped to handle the emotional challenges of language learning, resulting in higher enjoyment and sustained motivation [46, 47].

FLLE has a substantial impact on learning outcomes, particularly in terms of motivation, academic performance, and engagement. Studies consistently show that learners who experience greater enjoyment perform better academically and persist longer in their studies [29]. Positive emotional experiences encourage students to take risks, overcome challenges, and engage more fully in classroom activities [13, 48]. FLLE also helps reduce anxiety, a common barrier to language learning, by creating a positive emotional climate that makes the learning process more enjoyable and effective [49, 50]. However, the experience of FLLE can vary across cultural contexts. For instance, Jiang and Dewaele [6] found that Chinese EFL learners' experiences of FLLE were influenced by

cultural factors, such as exam-focused education and rote learning. Despite these differences, the core benefits of FLLE—enhanced motivation and improved learning—remain consistent across educational settings [13]. In traditionally challenging language learning environments, creating classrooms that foster enjoyment can significantly boost both engagement and performance [6].

In summary, FLLE is a crucial factor in shaping language learning experiences. Enjoyment from positive interactions with teachers and peers, as well as personal progress, plays a key role in fostering motivation, reducing anxiety, and improving academic outcomes. Understanding the factors that contribute to FLLE and its effects on learning can help develop more effective teaching strategies that create enjoyable, productive learning environments across different cultural contexts.

Motivated behavior in language learning

Motivated behavior serves as a cornerstone of success in SLA, intricately shaped by theoretical frameworks, individual characteristics, and environmental dynamics [2, 4]. Among the foundational theories, SDT emphasizes the distinction between intrinsic motivation, which is driven by personal satisfaction, and extrinsic motivation, focused on external rewards [5]. Intrinsic motivation often leads to sustained engagement and persistence, as learners derive satisfaction from the learning process itself [51]. In contrast, extrinsic motivation can result in inconsistent engagement, especially when learners prioritize external outcomes such as grades or social approval. Building on this, Dörnyei's [1] L2MSS introduces the constructs of the "ideal L2 self," which reflects a learner's aspirational self-image, and the "ought-to L2 self," rooted in external expectations. Empirical evidence demonstrates that a strong ideal L2 self is associated with higher engagement and academic achievement, underlining its critical role in driving motivated behavior [1, 29, 52]. Teimouri's [7] extension of the L2MSS further refines this understanding by presenting a trichotomous model, including the ideal L2 self, ought-to L2 self/own, and ought-to L2 self/others, which sheds light on how self-discrepancies influence learners' emotional responses and motivational outcomes [53].

Beyond these theoretical underpinnings, individual traits such as grit, resilience, and emotional intelligence significantly shape motivated behavior [46, 52, 54, 55]. Grit, defined as perseverance and passion for long-term goals, emerges as a powerful predictor of sustained effort and proficiency [20, 56]. Learners with higher levels of L2 grit tend to exhibit consistent study habits and greater resilience, allowing them to navigate challenges more effectively and achieve better language outcomes [7]. Resilience, characterized by traits like self-composure, sociability, and persistence, complements grit by enabling

learners to cope with setbacks and sustain motivation under pressure [57]. Emotional intelligence—the ability to recognize, understand, and manage emotions—further enhances learners' capacity to maintain motivation, fostering adaptability and reducing the risk of burnout [46, 54]. Studies illustrate how emotionally intelligent learners are better equipped to handle the emotional challenges of SLA, leading to increased persistence and engagement [58, 59]. Directed Motivational Currents (DMCs), which represent goal-oriented and sustained motivational states, exemplify the dynamic interplay between these individual traits, highlighting their role in fostering long-term engagement during intense learning efforts [59].

Environmental factors, particularly teacher support and classroom climate, also play a pivotal role in shaping motivated behavior. Teacher support, characterized by responsiveness and encouragement, consistently correlates with higher levels of engagement and sustained motivation among learners [30, 50, 57]. A classroom environment that emphasizes collaboration and interactive learning enhances motivation by fostering meaningful social interactions and a sense of belonging [43, 60, 59]. Within such environments, classroom practices that prioritize self-regulation and goal-setting are particularly effective. Self-regulated learners actively set attainable goals, monitor their progress, and reflect on their achievements, which helps them maintain motivation even in the face of challenges [61, 62]. These practices not only empower learners to take greater control of their academic journey but also contribute to their long-term success by mitigating the effects of obstacles and fostering resilience [10, 63, 64].

To encourage motivated behavior, educators must adopt strategies that address both individual and environmental factors. Prioritizing intrinsic motivation by fostering autonomy, competence, and relatedness—psychological needs highlighted by SDT—serves as a foundation for deeper engagement [5]. For instance, providing learners with opportunities to make meaningful choices, offering constructive feedback, and promoting social interactions can significantly enhance motivation. Simultaneously, supporting learners in developing self-regulation skills, such as goal-setting and progress monitoring, equips them to sustain motivation over time, especially when confronted with difficulties [63, 65]. Teacher encouragement remains equally critical, as a supportive and responsive teaching approach not only enhances learners' motivation but also creates a positive and engaging learning environment [66]. Additionally, integrating resilience-building activities, such as reflective exercises or mindfulness practices, can bolster learners' emotional intelligence and grit, reducing their vulnerability to burnout and enhancing their persistence in achieving long-term goals [67, 68, 69].

Overall, motivated behavior in SLA is a multifaceted construct influenced by a convergence of theoretical frameworks, individual characteristics, and environmental dynamics. The insights offered by SDT and L2MSS underscore the importance of intrinsic motivation, while the interplay between grit, resilience, and emotional intelligence highlights the value of individual perseverance and emotional regulation. Furthermore, the critical role of environmental factors such as teacher support and collaborative classroom climates reinforces the need for holistic approaches to fostering motivation. Practical strategies informed by these findings enable educators to create learning environments that promote sustained engagement, enhance language proficiency, and support learners' psychological well-being.

Academic burnout

Academic burnout, defined by emotional exhaustion, depersonalization, and a diminished sense of accomplishment, presents critical challenges to students' mental health, academic performance, and life satisfaction [22, 25, 54, 70, 67]. Although initially conceptualized in professional contexts during the 1970s, burnout has since been adapted to academic settings, where it manifests as fatigue stemming from academic demands, cynicism toward learning, and a diminished sense of efficacy [80, 73]. Theoretical models, such as Skinner and Pitzer's [74] framework, conceptualize burnout as the antithesis of engagement, where engaged students demonstrate energy and involvement, while those experiencing burnout withdraw and feel inadequate. Similarly, the Job Demand-Resource Theory posits that burnout occurs when academic demands, including intensive coursework and high-stakes exams, exceed the resources available to students, such as teacher support and institutional tools [75, 76]. These theoretical perspectives highlight the importance of institutional and interpersonal strategies to prevent and mitigate the adverse effects of burnout.

Burnout is driven by an interplay of environmental, personal, and motivational factors. Environmental stressors, including overcrowded classrooms, heavy academic pressure, and limited institutional support, are prominent contributors [28, 77]. Students who feel undervalued by their teachers or experience restricted autonomy in decision-making face heightened burnout risks [78]. At a personal level, traits like high anxiety, low resilience, and inadequate emotional regulation further exacerbate vulnerability [13, 79]. For instance, anxiety often leads to heightened emotional exhaustion, reducing students' capacity to stay engaged. Additionally, motivational orientations play a pivotal role in shaping burnout experiences. Students with mastery-oriented goals, focused on personal growth, tend to report lower levels of burnout compared to those driven by performance-oriented goals,

which are often associated with higher stress [73]. These findings underscore the protective potential of intrinsic motivation in buffering students against the pressures of academic life.

The consequences of burnout extend beyond academic performance, encompassing physical, psychological, and social dimensions. Physically, burnout manifests as chronic fatigue and stress-related health problems [77]. Psychologically, it is associated with increased levels of anxiety, depression, and feelings of hopelessness, which further undermine students' academic engagement [80]. Academically, burnout leads to absenteeism, reduced participation, and diminished achievement [26, 73]. Moreover, burnout negatively impacts life satisfaction, perpetuating a cycle in which academic stress spills over into personal well-being [26]. Research also demonstrates that cultural and demographic factors shape burnout experiences. Kuittinen and Meriläinen [81] found that educational practices and cultural expectations influence burnout differently across various cultural groups. Furthermore, factors like gender and academic discipline have been shown to affect students' susceptibility to burnout, highlighting the necessity of context-sensitive interventions.

Effectively addressing burnout requires a multifaceted approach that combines institutional and classroom-level strategies. At the institutional level, universities can mitigate burnout by offering accessible resources such as counseling services, academic advising, and structured stress management programs [75, 76]. Within classrooms, interventions such as fostering supportive learning environments, providing constructive feedback, and nurturing positive teacher-student relationships can play a crucial role in alleviating stress and enhancing student well-being [13, 50]. Moreover, encouraging self-regulation and resilience through goal-setting and time management has been shown to empower students to manage academic challenges more effectively [61, 62]. Practices like mindfulness exercises and physical activity further contribute to students' resilience and overall mental health, providing essential tools to combat the detrimental effects of burnout [74].

In EFL contexts, psychological and interpersonal factors play a particularly critical role in shaping burnout experiences. Psychological stressors, such as anxiety and apprehension, often impair students' ability to cope with academic pressures, exacerbating burnout symptoms [83]. Positive teacher-student relationships and enhanced enjoyment of the learning process can act as buffers against burnout, with emotional intelligence emerging as a key mediating factor [84]. For example, students with higher emotional intelligence are better equipped to manage their emotions, fostering resilience and reducing their susceptibility to academic stress. Furthermore,

supportive teacher behaviors and a positive classroom atmosphere provide the emotional scaffolding necessary to protect students from burnout's adverse effects.

Burnout in EFL learners is also influenced by demographic and cultural factors. High levels of academic stress and low institutional support are particularly common among senior high school students, resulting in elevated burnout rates [85]. However, research indicates that external and internal support mechanisms, such as peer collaboration and individual resilience, can effectively mitigate burnout and enhance engagement. Mixed-methods studies highlight the connection between specific burnout profiles and varying degrees of student engagement, underscoring the importance of tailored interventions that address the unique stressors faced by different demographic groups.

Resilience has been identified as a critical protective factor against burnout, supporting both motivated behavior and academic engagement. Resilient students are better prepared to navigate academic challenges and recover from setbacks, thereby reducing their risk of burnout [86]. Additionally, academic buoyancy—the ability to cope with everyday academic challenges—has been shown to mediate the relationship between social support and burnout [87]. These findings emphasize the importance of fostering resilience through practices that cultivate adaptability and perseverance. Moreover, a positive school climate and institutional policies that promote inclusivity and growth-oriented mindsets significantly lower burnout rates, further underscoring the impact of the broader learning environment [88].

Although considerable progress has been made in understanding academic burnout, several critical areas require further investigation to fully elucidate its impact on language learners. Although existing studies have explored the relationship between burnout and various factors, such as anxiety [83] and resilience [86], a more nuanced understanding of how burnout interacts with grit, learning enjoyment, and motivation in the context of language learning is needed. Specifically, further research is needed to investigate how positive emotional experiences, such as enjoyment in language learning, and perseverance towards long-term goals, as embodied in L2 grit, can mitigate the detrimental effects of burnout and foster sustained motivation. This line of inquiry has the potential to inform the development of pedagogical approaches that foster resilience, enhance motivation, and ultimately promote student well-being in academic contexts.

Grit in language learning contexts

Grit, defined as perseverance and passion for long-term goals, offers critical insights into academic success and personal achievement [15, 18]. It reflects sustained effort

and commitment to achieving meaningful objectives despite challenges [15, 77]. Grit consists of two dimensions: consistency of interest, which involves maintaining focus on long-term aspirations, and perseverance of effort, which refers to sustaining hard work over time. The Grit-S scale, widely used to measure these dimensions, has demonstrated robust predictive validity in domains such as education, job retention, and competitive performance [15, 17–19]. In educational settings, grit is linked to positive outcomes, including higher grades, extended study persistence, and increased program completion rates [117–119]. Meta-analytic research supports moderate but significant correlations between grit and academic performance, emphasizing its role in fostering resilience and enabling students to overcome academic challenges [20, 21]. This resilience allows learners to remain focused on their goals, even in demanding contexts such as universities and specialized training programs [19, 79].

Building on the general concept of grit, L2 grit has emerged as a key construct in SLA and EFL research, offering critical insights into learners' persistence and success in language learning [7, 93]. L2 grit emphasizes the sustained effort and commitment required to achieve language proficiency, addressing challenges unique to second language acquisition, such as overcoming communicative barriers and maintaining motivation through repetitive practice [7, 94]. Like its general counterpart, L2 grit comprises two dimensions: consistency of interest in the language learning process and perseverance of effort in language-related tasks [7]. These dimensions are measured using domain-specific tools such as the L2-Grit Scale, which has demonstrated strong predictive validity across diverse linguistic and cultural contexts [93, 95].

In SLA contexts, L2 grit is strongly associated with positive language learning outcomes. Studies have shown that learners with higher levels of L2 grit tend to engage more consistently in language practice, seize learning opportunities, and sustain motivation over time [55, 96, 93]. For example, Lake [55] observed that Japanese learners with higher L2 grit were more enthusiastic about studying English, while Changlek and Palanukulwong [96] reported that Thai students with high L2 grit experienced reduced language anxiety and greater motivation. Teimouri et al. [93] found that L2 grit significantly predicts language achievement, emphasizing the importance of sustained interest and effort in mastering a second language. These findings underscore L2 grit's indispensable role in maintaining the persistence necessary for language mastery.

Recent research has emphasized the need for a domain-specific conceptualization of grit in language learning. Sudina and Plonsky [94] demonstrated that while general grit overlaps with constructs like intended

effort, L2 grit remains distinct and is a stronger predictor of foreign language achievement. They highlighted that consistency of interest in the L2 is particularly crucial. Similarly, Teimouri, Sudina, and Plonsky [93] argued that tailored measures of L2 grit enhance predictive validity and provide nuanced insights into its effects across linguistic and cultural contexts. Cultural and contextual factors also shape the manifestation and impact of L2 grit. Sudina et al. [95] investigated the psychometric properties of the L2-Grit Scale and found differing correlations in EFL and ESL settings. For instance, perseverance of effort positively predicted proficiency in EFL contexts, whereas consistency of interest showed a negative association in ESL environments. These findings underscore the importance of culturally responsive approaches to fostering L2 grit. Similarly, Pawlak, Fathi, and Kruk [97] validated the Domain-Specific Grammar Grit Questionnaire across Iranian and Polish students, revealing significant cross-cultural differences that highlight how L2 grit is influenced by diverse educational practices and cultural norms.

The role of L2 grit as a mediator and moderator in language achievement has been explored in relation to other psychological constructs. Zhang and Zhang [98] demonstrated that L2 grit enhances EFL learners' writing performance through its mediation by cognitive abilities. Pawlak, Zarrinabadi, and Kruk [99] found that positive emotional experiences and high levels of L2 grit predict motivated behavior, emphasizing L2 grit's capacity to foster emotional resilience and consistent engagement. Fathi and Hejazi [100] identified L2 grit and foreign language enjoyment as mediators in the relationship between the ideal L2 self and language achievement, highlighting the synergy between intrinsic motivation and resilience in fostering success.

The dynamic relationship between L2 grit and emotional factors has also been investigated. Derakhshan and Fathi [101] explored the interactions among enjoyment, anxiety, willingness to communicate, and L2 grit, revealing that L2 grit both influences and is influenced by these emotional and motivational variables. Mohammad Hosseini et al. [102] identified mindfulness and L2 grit as key factors in reducing burnout and boredom, thereby enhancing academic engagement. These studies collectively highlight L2 grit's multifaceted role in sustaining motivation and promoting emotional well-being in language learning contexts.

Teacher support and the learning environment significantly affect L2 grit and related outcomes. Sadoughi and Hejazi [103] demonstrated that teacher support and growth-oriented mindsets positively influence academic engagement through L2 grit, underscoring the importance of nurturing educational settings. Paradowski and Jelińska [104] found that autonomy, curiosity,

and language mindsets predict L2 grit in online learning contexts, suggesting that fostering autonomy and curiosity can further enhance L2 grit's role in academic success. Finally, L2 grit's integration with other motivational constructs amplifies its impact on language performance. Fathi et al. [105] highlighted the combined effect of growth mindset, self-efficacy, and L2 grit in improving academic outcomes, while Fathi, Pawlak, and Hejazi [106] demonstrated that L2 grit, coupled with positive self-concepts and growth mindsets, is a significant predictor of writing success. These findings reinforce L2 grit's critical role as an enabler of resilience and sustained motivation in language learning.

Despite increasing research on L2 grit, key areas remain underexplored, particularly its role within the broader network of affective and motivational factors in language learning. While studies have linked L2 grit to variables like motivation, anxiety, and willingness to communicate, its potential as a mediator between positive emotions (e.g., enjoyment) and academic burnout requires further investigation. Addressing this gap could clarify the complex interplay among these factors and inform targeted pedagogical strategies to enhance resilience and long-term motivation in language learners.

Relationships among FLLE, motivated behavior, academic burnout, and grit

The interplay between FLLE, motivated behavior, academic burnout, and grit forms a framework that significantly impacts students' academic outcomes, emotional well-being, and resilience. FLLE plays a key role in fostering motivation and engagement by enhancing intrinsic motivation. Positive emotional experiences in language learning not only increase engagement but also lead to greater persistence, allowing students to actively participate, explore additional opportunities, and remain resilient when faced with challenges [8, 11]. These positive experiences also help buffer against emotional exhaustion, a major component of academic burnout, making learning more enjoyable and less taxing [9, 38].

On the other hand, academic burnout can severely undermine students' motivation and performance [22, 25]. FLLE helps mitigate burnout, as higher levels of enjoyment in learning alleviate stress and fatigue from academic demands [41, 107]. Additionally, supportive social interactions and classroom environments that nurture FLLE foster a sense of belonging, reducing detachment and cynicism often linked to burnout [44]. Grit sustains motivation and effort in academic contexts. Positive emotional experiences in language learning contribute to the development of grit, as students who find enjoyment in learning are more likely to remain committed and persevere toward academic goals despite obstacles [15, 55]. Promoting enjoyable learning experiences

can help students build grit, which in turn enhances their ability to persist in the face of academic challenges [19, 96].

Motivated behavior, marked by sustained effort and engagement, is inversely related to academic burnout. High levels of intrinsic motivation protect against burnout by encouraging proactive coping strategies and goal-setting behaviors that reduce the emotional toll of academic stress [29, 61]. Conversely, motivation driven by external pressures, such as grades or parental expectations, makes students more vulnerable to burnout, as external rewards increase stress and diminish personal satisfaction [5, 73]. Fostering intrinsic motivation in supportive environments is essential for reducing burnout and sustaining engagement in learning.

The relationship between grit and motivated behavior is also significant. Both traits involve sustained effort toward long-term goals, and students with higher grit levels tend to display more motivated behavior, driven by their passion and commitment [19, 20]. This reciprocal relationship suggests that success in learning reinforces long-term objectives, fostering perseverance and resilience [55, 96]. By cultivating both grit and motivated behavior, students can develop the resilience needed for academic success [79, 108].

Grit also serves as a buffer against burnout, as students with higher grit levels are more resilient and better equipped to manage academic stress. They are less likely to experience emotional exhaustion or disengagement [31, 109]. However, academic burnout can erode grit over time, diminishing students' passion and perseverance [22, 25]. Interventions that reduce burnout and promote positive emotional experiences are crucial for maintaining and enhancing grit, helping students stay focused on their long-term academic goals [77, 110].

While substantial research has explored the relationships among FLLE, motivated behavior, academic burnout, and grit, some areas require further investigation. Longitudinal studies are needed to examine how these variables interact over time and to identify key points for intervention [15, 20]. Additionally, cultural and contextual factors that shape these relationships should be explored, as educational systems and cultural norms may influence how students develop and express these traits [81]. Further research should also investigate the roles of psychological constructs such as emotional intelligence, self-efficacy, and growth mindset, which could offer deeper insights into how these variables contribute to academic success [46, 111, 112]. Understanding these complex interactions can guide educators in developing targeted strategies that support student well-being and long-term academic achievement.

The purpose of the study

This study aims to investigate the relationships among FLLE, academic burnout, L2 grit, and motivated behavior in EFL learners. Drawing on theoretical frameworks and empirical evidence, it explores how FLLE, a positive emotional experience in language learning, influences motivated behavior and academic burnout. The study also examines L2 grit as a mediating factor, affecting both motivation and burnout. Grounded in Fredrickson's broaden-and-build theory of positive emotions [11] and Duckworth's [15] concept of grit, this research aims to clarify the role of emotions, resilience, and motivation in SLA.

The study proposes four key hypotheses. First, FLLE is expected to positively influence motivated behavior, as learners who enjoy language learning tend to engage more and persist in their studies [8]. Second, FLLE is predicted to reduce academic burnout, with positive emotions buffering against stress and emotional exhaustion, key elements of burnout [38]. Third, L2 grit is hypothesized to mediate the effects of FLLE by promoting perseverance and long-term engagement, thereby reducing burnout and enhancing motivation [18]. Finally, academic burnout is expected to negatively impact motivated behavior, as burnout undermines emotional well-being and cognitive engagement, leading to reduced motivation [25].

In addition to quantitative analysis, this study includes a qualitative component that explores learners' experiences with FLLE, grit, motivation, and burnout through narrative inquiry. This mixed-methods approach provides a deeper understanding of the emotional and motivational dynamics of language learning, complementing the statistical data with personal insights. By integrating both quantitative and qualitative findings, the study aims to offer a comprehensive view of the emotional and motivational factors that contribute to success and well-being in language learners. Overall, this research contributes to the growing literature on language learning by demonstrating how positive emotional experiences like FLLE can foster motivation, mitigate burnout, and strengthen resilience through L2 grit. The findings will provide valuable insights for language educators on how to create supportive learning environments that enhance both academic performance and student well-being.

Methods

This study utilized an explanatory sequential mixed methods design [113], integrating both quantitative and qualitative approaches to examine the relationships among FLLE, motivated behavior, academic burnout, and L2 grit in EFL learners in mainland China. An explanatory sequential design was chosen to first quantitatively explore the relationships among key variables through a

cross-sectional survey, followed by qualitative methods to provide deeper contextual insights. In the quantitative phase, a cross-sectional survey design was employed, using SEM to test the hypothesized relationships between FLLE, burnout, motivation, and grit. In the qualitative phase, we conducted semi-structured interviews and collected digital narratives to provide deeper contextual insights into the statistical results. This approach allowed the qualitative findings to enrich and clarify the initial quantitative outcomes [114].

Participants and procedures

This study included 534 undergraduate English majors from universities in mainland China, selected through convenience sampling to ensure accessibility while capturing a wide range of institutional, regional, and demographic diversity. This approach enabled the inclusion of participants from various socioeconomic backgrounds, proficiency levels, and learning experiences, reflecting the broader population of EFL learners in mainland China. Although convenience sampling has inherent limitations in achieving full representativeness, efforts were made to enhance diversity by recruiting from major urban centers such as Beijing, Shanghai, and Guangzhou, which attract students from across the country.

The sample comprised 208 males (38.95%) and 326 females (61.05%), with ages ranging from 18 to 28 years ($M=22.6$, $SD=2.7$). Participants were required to meet specific eligibility criteria, including (1) enrollment in an undergraduate English program, (2) a minimum of three years of formal English education, and (3) intermediate to advanced proficiency, validated through standardized tests such as CET-4, CET-6, or IELTS. This ensured that participants had sufficient exposure to English learning environments and could provide reliable data on the constructs under investigation. Exclusion criteria included learners with less than three years of English education or those unable to provide informed consent.

Demographic data highlighted the diversity of the sample: 45.2% identified as middle-income, 35.4% as lower-middle-income, and 19.4% as upper-middle-income. Additionally, 28.6% had short-term exposure to English-speaking environments through cultural exchange programs, and 75% reported participation in supplementary English classes or tutoring. Participants' engagement with English-language media was also notable, with 68.7% regularly interacting with English movies, TV shows, and music. On a 10-point self-reported proficiency scale, participants' average score was 6.7 ($SD=1.1$), indicating intermediate to advanced proficiency.

Data collection was conducted over six months (March to August 2023) following ethical approval from the Institutional Review Board (IRB) at the lead researcher's university. Recruitment was carried out through email

invitations, announcements via language institutes, universities, and social media platforms popular among EFL learners. Efforts to enhance response rates included sending biweekly reminder emails.

For the qualitative phase, 15 participants were purposively selected from the initial sample based on their scores in FLLE, academic burnout, motivation, and L2 grit. This subset represented diverse experiences, ranging from high enjoyment and low burnout to high burnout and low motivation, ensuring a rich understanding of emotional and motivational dynamics in language learning. Purposive sampling was employed to maximize the depth and variety of insights, aligning the qualitative sample with the broader quantitative findings.

The study adhered to ethical standards throughout, ensuring informed consent, confidentiality, and secure data storage. By combining quantitative and qualitative approaches, this research aimed to provide a comprehensive exploration of emotional, motivational, and resilience factors influencing EFL learners.

Instruments

Quantitative instruments

Foreign Language Learning Enjoyment (FLLE) scale The enjoyment experienced by learners in their foreign language studies was measured using a 10-item scale adapted from the original questionnaire by [8], as utilized by Jiang and Dewaele [6]. This scale conceptualizes FLLE as comprising two dimensions: Social Enjoyment (5 items, e.g., "My peers in the English class are nice.") and Private Enjoyment (5 items, e.g., "I feel proud of my accomplishments in this term's English class."). Social Enjoyment reflects positive emotional interactions in social contexts, while Private Enjoyment emphasizes personal satisfaction and achievements. Participants rated items on a 5-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). This scale has been validated in diverse EFL contexts, demonstrating strong psychometric properties across different learner populations [6, 100, 115]. In this study, the scale demonstrated high internal consistency (Cronbach's $\alpha=0.88$). Construct validity was confirmed through CFA, with fit indices meeting established benchmarks (CFI=0.96, TLI=0.95, RMSEA=0.045), supporting the scale's suitability for this population.

L2 grit scale Participants' L2 grit was assessed using the scale developed and validated by Teimouri et al. [7]. This instrument consists of 9 items divided into two subscales: Consistency of Interest (4 items, e.g., "My interests in learning English change from year to year.") and Perseverance of Effort (5 items, e.g., "I put much time and effort into improving my English weaknesses."). Participants

responded on a five-point Likert scale. Previous research has consistently supported the scale's psychometric soundness, with applications in various educational and cultural settings [7, 102, 116]. In the current study, the scale exhibited strong internal consistency (Cronbach's $\alpha=0.83$). CFA results confirmed construct validity, yielding CFI=0.94, TLI=0.93, and RMSEA=0.048, ensuring the scale's appropriateness for capturing L2 grit among EFL learners.

Motivated behavior scale Motivated behavior was evaluated using 10 items from the study by Taguchi et al. [117], which measure learners' willingness to invest effort and time in studying English (e.g., "I would like to spend lots of time studying English"). Responses were recorded on a 5-point Likert scale. The scale has demonstrated reliability and validity in prior cross-cultural studies involving EFL learners [117, 99]. In this study, the scale showed excellent reliability (Cronbach's $\alpha=0.86$). The CFA results further confirmed its construct validity, with CFI=0.95, TLI=0.94, and RMSEA=0.049. These results underscore the scale's robustness in assessing motivated behavior within diverse EFL contexts.

Academic burnout scale Academic burnout was measured using a scale validated by Li et al. [13], based on the Maslach Burnout Student Survey developed by Schaufeli et al. [23]. The scale encompasses three subscales: Exhaustion (4 items, e.g., "I feel burned out from my English studies"), Cynicism (3 items, e.g., "I have become less enthusiastic about my English studies"), and Reduced Efficacy (3 items, e.g., "During English class, I don't feel that I am effective in getting things done"). This instrument has been validated in academic contexts, particularly in language learning, demonstrating strong reliability and validity across populations [13, 73]. In this study, the scale achieved excellent reliability (Cronbach's $\alpha=0.91$). CFA results further supported the scale's psychometric integrity, with indices of CFI=0.93, TLI=0.92, and RMSEA=0.052, indicating its robustness in capturing academic burnout among EFL learners.

Qualitative instruments

To explore participants' experiences in depth, we employed two complementary qualitative methods: semi-structured interviews and digital narratives. This approach aligns with Creswell and Creswell's [113] recommendations for using multiple qualitative data collection strategies to enrich understanding.

Semi-structured interviews The semi-structured interviews involved direct, one-on-one conversations between the researcher and each participant. This method allowed for interactive dialogue, where participants could share

detailed accounts of their experiences with enjoyment, motivation, burnout, and grit in learning English. The semi-structured format provided a balance between guided questioning and open-ended responses, enabling participants to express themselves freely while allowing the researcher to probe deeper into specific areas of interest. Participants were asked open-ended questions, such as:

- "Can you tell me about a time when you felt the most engaged and motivated in your English studies? Please share a specific example that highlights what made this experience motivating for you."
- "What experiences have contributed to your perseverance in learning English despite challenges or setbacks? Can you describe an instance where your perseverance helped you overcome a particular obstacle?"
- "Can you describe any periods where you felt burned out or less enthusiastic about continuing your English learning journey? Please provide an example of what triggered these feelings and how you responded to them."

These interviews were conducted with the purposive sample of 15 participants. Each interview lasted approximately 45–60 min and was audio-recorded for later transcription and analysis.

Digital narratives In addition to the interviews, participants were asked to create digital narratives—self-recorded video or audio stories—in which they reflected on their English learning journeys. The digital narratives offered participants the opportunity to independently articulate their experiences, thoughts, and feelings without the immediate presence of the researcher. They were provided with prompts to guide their reflections on themes such as moments of enjoyment, challenges faced, perseverance through difficulties, and experiences of burnout. However, they had the flexibility to share their stories in their own words and style. Prompts included:

- "Describe a moment when you felt the most motivated or engaged in your English studies."
- "Can you recall any significant challenges you encountered while learning English? How did you overcome them?"
- "Reflect on a time when you experienced burnout or lost motivation. What helped you regain your focus?"

Although both methods aimed to elicit rich, personal narratives, they differed in their nature and interaction level. The semi-structured interviews were interactive and allowed for real-time dialogue and clarification,

aligning with qualitative interview techniques [113]. The researcher could probe deeper based on participants' responses, facilitating a dynamic exploration of their experiences. In contrast, the digital narratives were participant-led and reflective, allowing participants to control the content and pacing of their storytelling without direct researcher influence during the recording. This distinction provided a balance between guided exploration and independent expression, enriching the qualitative data with diverse perspectives [113].

Data analysis

Data analysis involved both quantitative and qualitative approaches, ensuring a comprehensive understanding of the relationships between key variables. Quantitative data were analyzed using IBM SPSS Statistics (Version 27) and AMOS (Version 26) for SEM. These software packages were chosen for their robust capabilities in statistical analysis and their widespread use in educational and psychological research. SPSS was used for preliminary analyses, including descriptive statistics and reliability assessments, while AMOS was employed for the advanced SEM analyses, which allow for the testing of complex relationships among latent and observed variables.

The analysis began with descriptive statistics (means, standard deviations, and frequency distributions) to summarize demographic characteristics and the distribution of key variables. Reliability assessments were conducted using Cronbach's alpha to determine the internal consistency of the scales used. All scales demonstrated good reliability, with the FLLE scale yielding a Cronbach's alpha of 0.87, the L2 Grit scale 0.828, the Motivated Behavior scale 0.86, and the Academic Burnout scale 0.91.

Pearson correlation analyses were performed to assess the strength and direction of relationships between FLLE, motivated behavior, academic burnout, and L2 grit. These preliminary analyses guided the subsequent SEM analysis, which tested the hypothesized model positing that FLLE would have direct effects on motivated behavior and academic burnout, with L2 grit serving as a mediating variable. SEM allowed for the examination of complex relationships among multiple variables and provided a comprehensive test of the hypothesized pathways [91].

Model fit was assessed using several fit indices, including the chi-square statistic (χ^2), Comparative Fit Index (CFI), Tucker-Lewis Index (TLI), and Root Mean Square Error of Approximation (RMSEA). Both CFI and TLI values exceeded the recommended threshold of 0.90, indicating good model fit, while the RMSEA value was below 0.08, further supporting the model's adequacy [120, 93]. Bootstrapping methods with 5,000 samples were used

to estimate indirect effects and generate 95% confidence intervals for mediation effects [121].

For the qualitative analysis, the semi-structured narrative interviews and digital narratives were transcribed and analyzed using thematic analysis [114]. The analysis adhered to Braun and Clarke's [122] six-phase framework: familiarizing with the data, generating initial codes, identifying themes, reviewing and refining themes, defining and naming themes, and compiling the report. This structured process ensured a comprehensive and accurate representation of participants' experiences. An inductive coding approach was employed to allow themes to emerge directly from the data without being influenced by preconceived frameworks. Two researchers independently coded the transcripts, compared their findings, and resolved discrepancies through discussion to achieve consensus. A reflexive approach was incorporated to minimize bias, with researchers critically reflecting on their potential influence during the coding and analysis.

The analysis identified recurring themes related to enjoyment, motivation, burnout, and grit. These themes were organized hierarchically, with subthemes providing more detailed insights. For instance, within the theme of "enjoyment," subthemes such as "teacher-supported enjoyment" and "peer-supported enjoyment" highlighted specific sources of positive learning experiences. This thematic structure offered a nuanced understanding of learners' emotional and motivational dynamics, contextualizing the quantitative findings.

To ensure rigor, the thematic findings were triangulated with the quantitative results, validating and enriching the overall interpretation. Qualitative data analysis software (NVivo 12) was used to manage and systematically examine the data, enhancing transparency and replicability in the analytical process.

Results

Quantitative results

The quantitative results are presented in three sections: descriptive statistics, correlation analysis, and SEM analysis, offering a comprehensive view of the relationships between FLLE, motivated behavior, academic burnout, and the mediating role of L2 grit among EFL learners.

Descriptive statistics

Table 1 summarizes the descriptive statistics for the key variables, including means, standard deviations, skewness, and kurtosis. These statistics offer foundational insight into the participants' self-reported levels of FLLE, motivated behavior, academic burnout, and L2 grit.

As shown in Table 1, participants reported moderately high levels of FLLE ($M=3.76$, $SD=0.58$) and motivated behavior ($M=4.12$, $SD=0.67$). The mean score for

Table 1 Descriptive statistics with skewness and Kurtosis

Variable	Mean (M)	Standard Deviation (SD)	Skewness	Kurtosis
FLLE	3.76	0.58	-0.32	-0.45
Motivated Behavior	4.12	0.67	-0.27	-0.36
Academic Burnout	2.34	0.72	0.48	-0.12
L2 Grit	3.95	0.61	-0.22	-0.58

Table 2 Correlation matrix for key variables

Variable	1	2	3	4
1. FLLE	1			
2. Motivated Behavior	0.62**	1		
3. Academic Burnout	-0.48**	-0.55**	1	
4. L2 Grit	0.54**	0.65**	-0.46**	1

Note. $P < .01$

academic burnout was lower ($M=2.34$, $SD=0.72$), indicating less prevalence of burnout among the participants. L2 grit also showed a relatively high mean ($M=3.95$, $SD=0.61$), suggesting that participants demonstrated perseverance and passion for language learning.

The skewness and kurtosis values provide additional information about the distribution of these variables. Skewness values close to zero indicate a near-normal distribution, with slight negative skewness for FLLE, motivated behavior, and L2 grit, suggesting a slight concentration of higher scores. Academic burnout, on the other hand, exhibited a positive skew (0.48), indicating that lower burnout levels were more common. The kurtosis values for all variables are close to zero, showing no significant departure from normality. This suggests that the variables were approximately normally distributed, further validating the use of parametric statistical methods in subsequent analyses.

Correlation analysis

The Pearson correlation coefficients were calculated to examine the relationships between the study’s key variables, with the results presented in Table 2.

The correlation analysis revealed several significant relationships. FLLE was positively correlated with motivated behavior ($r=.62$, $p<.01$) and L2 grit ($r=.54$, $p<.01$), while it was negatively correlated with academic burnout ($r=-.48$, $p<.01$). These results suggest that higher enjoyment in language learning is associated with greater motivation and perseverance (grit), and lower levels of burnout. Similarly, motivated behavior was positively correlated with L2 grit ($r=.65$, $p<.01$) and negatively correlated with academic burnout ($r=-.55$, $p<.01$), indicating that higher motivation is related to both higher grit and lower burnout. Lastly, L2 grit was negatively correlated with academic burnout ($r=-.46$, $p<.01$), suggesting

Table 3 SEM path coefficients

Path	Standardized Estimate (β)	Standard Error (SE)	p-value
FLLE $\rightarrow$ Motivated Behavior	0.42	0.05	< 0.001
FLLE $\rightarrow$ Academic Burnout	-0.38	0.04	< 0.001
FLLE $\rightarrow$ L2 Grit	0.51	0.06	< 0.001
L2 Grit $\rightarrow$ Motivated Behavior	0.39	0.05	< 0.001
L2 Grit $\rightarrow$ Academic Burnout	-0.35	0.04	< 0.001
Academic Burnout $\rightarrow$ Motivated Behavior	-0.43	0.05	< 0.001

that students with more grit are less likely to experience burnout.

SEM analysis

In this study, we used covariance-based SEM (CB-SEM) to investigate both the direct and indirect effects of Foreign Language Learning Enjoyment (FLLE) on motivated behavior and academic burnout, with L2 grit serving as a mediating variable. CB-SEM was selected due to its strong capacity for theory testing and its ability to estimate complex relationships among latent constructs while accounting for measurement error.

Before proceeding with the SEM analysis, we evaluated whether key assumptions were met, including multivariate normality, linearity, and the absence of multicollinearity. Multivariate normality was assessed using Mardia’s test, which indicated a slight deviation (skewness=2.13, kurtosis=5.72). Although not perfectly normal, the deviation was minor, and SEM is generally robust to such violations. Linearity was confirmed by inspecting scatterplots of the residuals, which suggested that the relationships between latent variables were sufficiently linear. Multicollinearity was assessed using Variance Inflation Factor (VIF) values, all of which were below the threshold of 5, indicating no significant multicollinearity among predictors.

Given the slight deviation from multivariate normality indicated by Mardia’s test, we applied the Satorra-Bentler correction to adjust the chi-square statistic. This ensured that our model fit indices remained accurate, even in the presence of non-normal data distributions.

The SEM results, summarized in Table 3 and detailed in Fig. 1, indicate that the hypothesized model provided a good fit for the data. Specifically, the chi-square statistic was $\chi^2(84)=112.45$, $p=.018$, which is acceptable given the sample size. Additionally, the model demonstrated strong fit indices, with a CFI of 0.96 and a TLI of 0.95, both exceeding the recommended threshold of 0.90 [120]. The RMSEA was 0.045, falling below the cutoff of 0.08, further supporting the adequacy of the model fit [93].

The SEM analysis revealed that FLLE had a significant positive direct effect on motivated behavior ($\beta=0.42$, $SE=0.05$, $p<.001$) and a significant negative direct effect

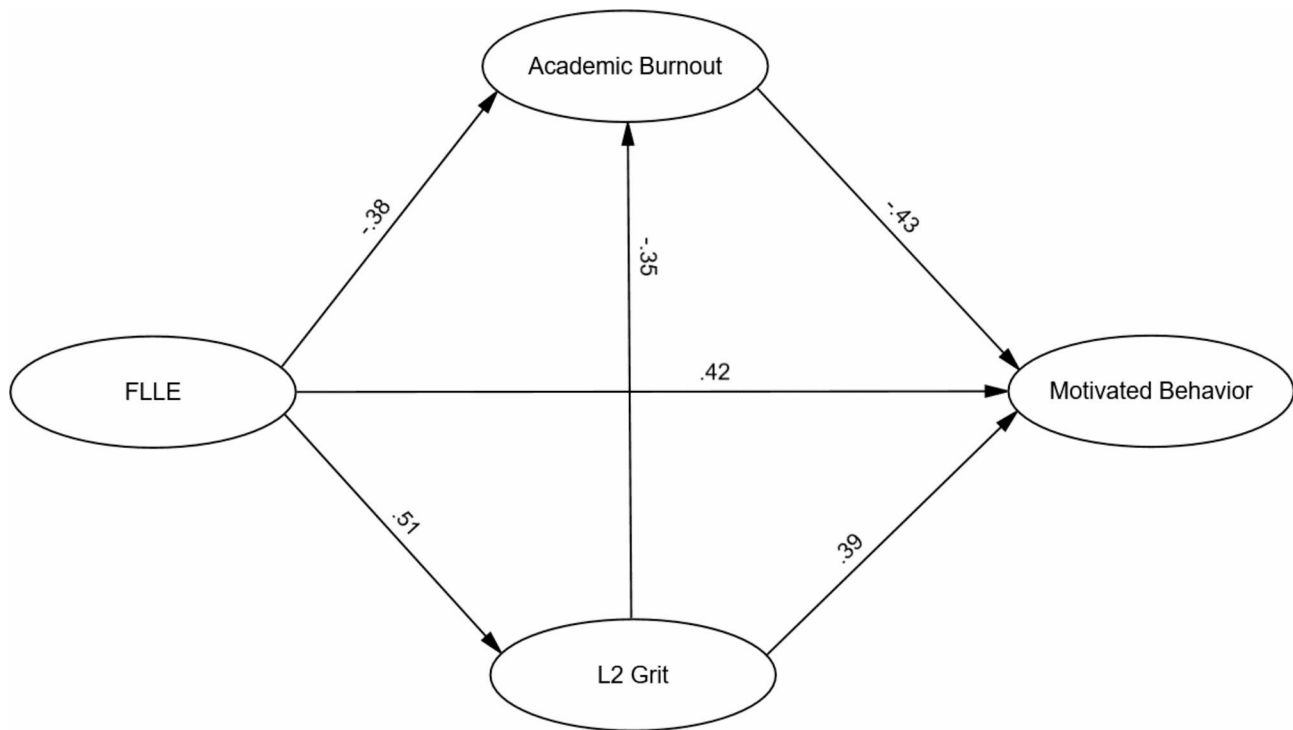


Fig. 1 The structural model

Table 4 Indirect effects of FLLE on motivated behavior and academic burnout

Indirect Path	Standardized Estimate (β)	Bootstrapped 95% CI	p-value
FLLE → Academic Burnout → Motivated Behavior	0.16	[0.10, 0.23]	< 0.001
FLLE → L2 Grit → Motivated Behavior	0.20	[0.13, 0.27]	< 0.001
FLLE → L2 Grit → Academic Burnout → Motivated Behavior	0.077	[0.04, 0.12]	< 0.001

on academic burnout ($\beta = -0.38$, $SE = 0.04$, $p < .001$). Additionally, FLLE significantly positively affected L2 grit ($\beta = 0.51$, $SE = 0.06$, $p < .001$). In turn, L2 grit significantly positively affected motivated behavior ($\beta = 0.39$, $SE = 0.05$, $p < .001$) and had significant negative effects on academic burnout ($\beta = -0.35$, $SE = 0.04$, $p < .001$). Furthermore, academic burnout significantly negatively impacted motivated behavior ($\beta = -0.43$, $SE = 0.05$, $p < .001$), indicating that as burnout increases, motivated behavior decreases.

Indirect effects

To further explore the relationships among FLLE, motivated behavior, and academic burnout, bootstrapping procedures with 5,000 resamples were employed to assess the indirect effects of FLLE through L2 grit and academic burnout. The results of the indirect effects are presented in Table 4.

The indirect effect of FLLE on motivated behavior through academic burnout was calculated by multiplying the standardized path coefficients of FLLE to academic burnout ($\beta = -0.38$, $p < .001$) and academic burnout to motivated behavior ($\beta = -0.43$, $p < .001$). The resulting indirect effect was $\beta = 0.16$. The bootstrapped 95% confidence interval (CI) for this effect was [0.10, 0.23], confirming its significance ($p < .001$). Similarly, the indirect effect of FLLE on motivated behavior through L2 grit was calculated by multiplying the path coefficients of FLLE to L2 grit ($\beta = 0.51$, $p < .001$) and L2 grit to motivated behavior ($\beta = 0.39$, $p < .001$). The resulting indirect effect was $\beta = 0.20$, with a bootstrapped 95% CI of [0.13, 0.27] ($p < .001$).

Finally, the indirect effect of FLLE on motivated behavior through both L2 grit and academic burnout was calculated by multiplying the path coefficients of FLLE to L2 grit ($\beta = 0.51$, $p < .001$), L2 grit to academic burnout ($\beta = -0.35$, $p < .001$), and academic burnout to motivated behavior ($\beta = -0.43$, $p < .001$). The resulting indirect effect was $\beta = 0.077$, with a bootstrapped 95% CI of [0.04, 0.12] ($p < .001$).

Among the three indirect pathways, the most significant path was FLLE → L2 Grit → Motivated Behavior ($\beta = 0.20$, $p < .001$). This pathway highlights the pivotal role of L2 grit in mediating the positive relationship between FLLE and motivated behavior, with the largest contribution to explaining the variance in motivation. The second most significant path was FLLE → Academic

Burnout $\rightarrow$ Motivated Behavior ($\beta=0.16$, $p<.001$). This pathway underscores the protective effect of FLLE in reducing academic burnout, which subsequently enhances motivated behavior. The least significant path was FLLE $\rightarrow$ L2 Grit $\rightarrow$ Academic Burnout $\rightarrow$ Motivated Behavior ($\beta=0.077$, $p<.001$). Although smaller in magnitude, this pathway demonstrates the nuanced interplay among FLLE, L2 grit, and academic burnout in shaping motivated behavior, with both grit and reduced burnout working synergistically to boost motivation.

To better understand the explanatory power of the model, the coefficient of determination (R^2) values for key dependent variables were analyzed. The R^2 for academic burnout was 0.28, indicating that 28% of the variance in burnout could be explained by FLLE and L2 grit. Similarly, the R^2 for motivated behavior was 0.42, suggesting that 42% of the variance in motivation was accounted for by FLLE, academic burnout, and L2 grit. These values indicate a moderate to substantial level of explanatory power, underscoring the robustness of the model in capturing the relationships among these variables.

The bootstrapping results confirm the mediating role of L2 grit and academic burnout in the relationship between FLLE and motivated behavior, emphasizing the complexity of the interrelationships among these variables. Specifically, FLLE reduces academic burnout, which in turn enhances motivated behavior, while L2 grit both directly strengthens motivation and mitigates burnout. The inclusion of R^2 further reinforces the model's validity by quantifying the extent to which the independent variables collectively explain the variance in the dependent variables.

Qualitative results

The qualitative phase utilized semi-structured narrative interviews and digital narratives to explore learners' experiences of FLLE, perseverance, and burnout, addressing RQ3: How do learners' personal experiences of enjoyment, perseverance, and burnout provide deeper insights into emotional and motivational dynamics in language learning? Thematic analysis revealed several interconnected themes, highlighting the complex emotional and motivational dynamics that underpin language learning. Participants are referred to using pseudonyms to maintain confidentiality.

The role of enjoyment in sustaining motivation

A central theme emerging from the narratives was the significance of enjoyment in fostering sustained motivation. Positive learning experiences acted as a catalyst, reenergizing learners during stressful periods and enabling them to maintain their motivation. Li Wei, a 21-year-old English major from Beijing, reflected: "There was a time when I felt really stressed out, especially

with all the assignments piling up. But in one class, we watched a movie without subtitles, and I could follow along almost perfectly. That gave me so much confidence—it was fun and made everything feel worth it."

This account demonstrates how enjoyable experiences not only reduced the cognitive burden of language learning but also acted as a psychological buffer against stress. Similarly, Xiaohong, a student from Guangzhou, shared: "The first time I understood a whole episode of my favorite English show without needing subtitles, I felt like I was actually getting somewhere. That sense of achievement kept me motivated to keep learning."

These narratives illustrate the powerful role of intrinsic enjoyment in driving engagement and persistence, aligning with Fredrickson's broaden-and-build theory, which posits that positive emotions enhance cognitive and motivational resources. Enjoyment emerged as a vital element, enabling learners to sustain motivation during demanding phases of their learning journey.

Perseverance in the face of setbacks (L2 grit)

Perseverance, as an expression of L2 grit, was another critical theme, with participants highlighting its role in overcoming setbacks and sustaining long-term commitment. For instance, Chen Rui, a student from a rural area of Hunan, described how repeated failures motivated him to work harder: In the beginning, I was always behind in class, especially in pronunciation. It was frustrating, but I kept practicing at home. Over time, I started seeing small improvements, and that pushed me to keep going. It wasn't easy, but I've learned that progress takes time.

Similarly, Min Jia recounted how a significant setback—failing an important exam—became a turning point: "Failing made me doubt whether I was really cut out for learning English. But then I thought about how much I'd already invested and decided I couldn't quit. Slowly, I picked myself back up and worked even harder."

These stories underscore the role of grit in maintaining perseverance despite difficulties. The learners' reflections demonstrate how setbacks, when coupled with a commitment to long-term goals, serve as opportunities for growth, fostering resilience and sustained motivation.

The emotional impact of burnout

Burnout, often tied to intense academic demands, emerged as a recurring experience among participants. Emotional exhaustion was frequently cited as a barrier to motivation and engagement. Yue Lin, an advanced learner from Shanghai, vividly described her experience of burnout:

"During the exam period, I felt completely drained. No matter how much I studied, it never felt enough. I started to wonder if all this hard work was even worth it because I wasn't seeing any improvement. It was like hitting a

wall.” Zhang Lei shared a similar sentiment: “I studied for hours every day, but eventually, it just felt like I wasn’t moving forward. I started losing motivation because I couldn’t see the point anymore.” These narratives highlight how burnout can lead to a cycle of disengagement, where emotional exhaustion undermines confidence and motivation.

The protective role of positive learning experiences

Despite the challenges posed by burnout, positive learning experiences emerged as a protective factor, helping learners regain motivation and confidence. Liang Mei, a student from a low-income background, described a moment that renewed her determination: “There was one time I understood a really difficult reading passage, something I had been struggling with for weeks. That moment gave me so much confidence—it reminded me that I was capable and that my efforts were paying off. It helped me push through when I was feeling really down.” Wang Jian emphasized the importance of teacher feedback: “When my teacher told me that my pronunciation had improved, I felt like my hard work was finally being recognized. It wasn’t a huge improvement, but just hearing that made me want to keep going.”

These experiences demonstrate how moments of achievement and constructive feedback mitigate the negative effects of burnout, fostering resilience and persistence.

Interplay between intrinsic and extrinsic motivation

Participants often distinguished between intrinsic and extrinsic motivators, noting that while external pressures (e.g., exams and parental expectations) drove initial engagement, intrinsic motivation ultimately sustained long-term effort. Chen Yan reflected on this transition: At first, I studied English just to get good grades. But after a while, I started enjoying reading books in English for fun. The pressure from exams faded, but my interest stayed.

In contrast, Hao Ming described how extrinsic motivators were insufficient to sustain engagement:

“Before exams, I was highly motivated because I wanted to pass. But after the exams, I lost interest. It wasn’t until I found English movies that I enjoyed that my motivation came back.” These narratives highlight the interplay between extrinsic and intrinsic motivators, with the latter proving essential for long-term engagement and emotional resilience.

Overall, the qualitative findings align closely with RQ3, offering rich insights into how learners’ experiences of enjoyment, perseverance, and burnout shape their emotional and motivational dynamics in language learning. Positive emotional experiences foster motivation and resilience, while grit enables learners to persist through

setbacks. Burnout, though challenging, can be mitigated by supportive feedback and moments of achievement, reinforcing the protective role of positive learning environments. These insights provide valuable implications for fostering emotionally supportive and motivationally enriching language learning experiences.

Discussion

This study investigated the relationships among FLLE, academic burnout, L2 grit, and motivated behavior in EFL learners from mainland China. By employing a mixed-methods approach that combined SEM analysis with qualitative data from narrative interviews and digital stories, the research provided a comprehensive understanding of how emotional experiences, motivational factors, and resilience contribute to academic success and well-being in language learning.

Concerning the first research question, it was revealed that FLLE has a direct positive effect on motivated behavior

The SEM results indicated that enjoyment fosters intrinsic motivation, highlighting how positive emotional experiences enhance engagement and persistence. This aligns with prior research [6, 8, 10, 11, 52, 67], as well as recent studies [84, 99, 115], which demonstrate that learners who experience positive emotions are more likely to develop intrinsic motivation. For instance, Li and Zhang [84] found that enhanced learning enjoyment, mediated by emotional intelligence, significantly reduces burnout among EFL students. Similarly, Pawlak et al. [99] showed that positive emotions positively predict L2 motivated behavior, and Fathi et al. [115] identified FLLE as a significant predictor of willingness to communicate. These effects can be attributed to intrinsic motivation, driven by internal satisfaction and genuine interest in the task [5]. When learners enjoy the language learning process, they are more inclined to invest effort, as the activity becomes inherently rewarding [12, 123]. Qualitative data supported these findings, with participants noting that activities such as joining speaking clubs or celebrating personal milestones reignited their motivation during challenging periods. These experiences resonate with Fredrickson’s broaden-and-build theory [11], suggesting that positive emotions expand cognitive resources, enabling learners to remain engaged and persevere in the face of obstacles [38].

With regard to the second research question, it was found that FLLE significantly reduces academic burnout, highlighting its protective role against emotional exhaustion, cynicism, and inefficacy

This is consistent with existing research on burnout [22, 23, 61, 70], including recent studies in EFL contexts [83–86]. For example, Li and Zhang [84] emphasized that positive teacher–stu-

dent dynamics and increased learning enjoyment mitigate burnout, with emotional intelligence serving as a key mediating factor. Wu et al. [85, 86] reported that burnout among Chinese EFL learners is closely linked to engagement levels, and that resilience can buffer its negative effects. Positive emotions generated through enjoyment help alleviate the stress of demanding academic environments, enabling students to manage challenges more effectively [41]. This protective effect likely stems from the ability of positive emotions to build psychological resilience, aiding learners in coping with stress and preventing burnout [124, 110]. Qualitative insights reinforced this, with participants describing how small achievements or positive feedback from instructors rejuvenated their energy and provided emotional relief. These findings underscore the importance of fostering supportive and emotionally positive learning environments to enhance motivation and safeguard student well-being in stressful academic contexts [25, 70]. When educators integrate enjoyment and positive reinforcement into their teaching methods, students are better equipped to handle stress and maintain strong academic performance [13, 78]. Additionally, studies by Fu [87] and Liu [88] highlight the crucial role of social support and a positive school climate in reducing burnout among EFL learners.

As for the third research question, we found that L2 grit mediates the relationships among FLLE, motivated behavior, and burnout The SEM analysis indicated that FLLE positively influences L2 grit, which in turn enhances motivated behavior and reduces burnout. This supports the notion that perseverance and passion for long-term goals are vital for sustaining motivation and mitigating burnout, aligning with Duckworth et al.'s [15] conceptualization of grit. Recent research in SLA emphasizes the significance of domain-specific grit in language learning [53, 94, 93]. Teimouri et al. [93], for instance, advocated for a domain-specific understanding of grit, finding that L2 grit significantly predicts language achievement. Sudina and Plonsky [94] demonstrated that language-specific grit is a strong predictor of academic perseverance in foreign language learning. Grit functions as a motivational resource, enabling learners to persist through difficulties by reinforcing their commitment to long-term objectives [7, 18, 80, 111]. FLLE likely enhances grit by boosting enthusiasm and sustaining effort over time. Qualitative data provided further insight, with participants sharing how grit, often fueled by enjoyment, helped them overcome repeated failures and challenges. One participant described persisting through multiple exam failures driven by the goal of mastering English, while others expressed determination to continue despite feeling less naturally skilled, illustrating how grit fosters resilience and persistence, effectively reducing the impact

of academic stress [20, 31, 56]. Studies by Pawlak et al. [99] and Fathi and Hejazi [100] further support these findings, showing that L2 grit mediates the relationship between positive emotions and motivated behavior.

The strong negative relationship between academic burnout and motivated behavior ($\beta = -0.43, p < .001$) observed in the SEM analysis indicates that emotional exhaustion significantly diminishes motivation. This finding corroborates previous research [22, 25, 110], including recent EFL studies [83, 86, 87], which highlight how burnout erodes intrinsic motivation and engagement, leading to lower academic performance. Ren [83] found that apprehension and anxiety substantially contribute to burnout among EFL students, negatively affecting their motivation. Wu et al. [86] demonstrated that burnout adversely impacts engagement, with resilience playing a critical role in mitigating this effect. The qualitative data echoed these results, as participants expressed feelings of exhaustion and inefficacy that drained their motivation to continue studying. These findings emphasize the detrimental impact of burnout on both emotional well-being and academic outcomes [24, 26, 61]. Addressing these issues necessitates implementing strategies aimed at reducing burnout in educational settings, such as stress management programs, enhancing social support, or encouraging self-care among students [27, 28, 87, 102]. Reducing burnout is essential for maintaining motivation and ensuring long-term academic and emotional well-being [67, 78].

The transition from extrinsic to intrinsic motivation emerged as a critical factor in sustaining long-term engagement. Several participants noted that while their initial motivation to study English was driven by external factors like exams or parental expectations, they eventually developed intrinsic motivation through personal enjoyment in activities such as reading English books or watching English-language media. This shift aligns with Self-Determination Theory [5] and the L2 Motivational Self System [1, 66, 69], which emphasize the role of the ideal L2 self in fostering intrinsic motivation. Khany and Amiri [66] found that the ideal L2 self significantly predicts motivated learning behavior, and Kim and Kim [69] demonstrated its positive influence on motivated behavior and English proficiency. Fathi et al. [125] further explored the relationships among FLLE, ideal L2 self, grit, and growth mindset, highlighting the importance of intrinsic motivational factors for sustained engagement. Intrinsic motivation is sustainable because it is driven by the inherent satisfaction of the task rather than external pressures, making it a more reliable source of long-term engagement [1, 126, 96]. These findings underscore the need to promote enjoyment and personal interest in language learning, as fostering intrinsic motivation can lead

to more resilient, self-regulated learners capable of sustained effort over time [30, 61, 67].

Overall, the integration of quantitative and qualitative findings highlights the interconnected roles of FLLE, grit, and burnout in shaping motivation and academic resilience. The results suggest that by encouraging positive emotional experiences and fostering perseverance, educators can enhance student engagement, reduce burnout, and support long-term success in language learning. Recent research supports these conclusions, emphasizing the importance of these factors. For instance, Kim and Kim [57] found that resilience and L2 motivation significantly impact motivated behavior and proficiency. Yu and Liu [59] highlighted the dynamic nature of motivation and emotions over time in L2 learning, emphasizing the need for longitudinal research. Longitudinal studies by Derakhshan and Fathi [101] and Fathi et al. [106] have explored the complex interactions among enjoyment, grit, motivation, and achievement, underscoring the importance of understanding how these factors evolve. Future research should continue to investigate these relationships through longitudinal studies to explore changes over time. Examining cultural and contextual factors that influence these dynamics is also crucial, as cross-cultural studies like those by Pawlak et al. [97] can provide insights for developing tailored interventions in diverse educational environments.

Conclusion

This study offers significant contributions to understanding the interplay between emotional and motivational factors in language learning, specifically highlighting the relationships among FLLE, academic burnout, L2 grit, and motivated behavior. The findings underscore the central role of FLLE in promoting sustained motivation and resilience among EFL learners, with positive emotional experiences in language learning acting as both a driver of engagement and a buffer against burnout. The discovery that L2 grit mediates the effects of FLLE on both motivated behavior and academic burnout further emphasizes the importance of fostering perseverance and passion for long-term language goals. Ultimately, this research enriches the field by demonstrating how emotional and motivational factors can support learners not only in achieving academic success but also in maintaining well-being throughout the language acquisition process. The broader implications of these findings are relevant to both pedagogical practice and institutional policy, suggesting that fostering positive emotional climates and resilience are essential for learners' long-term success in acquiring a second language.

Implications

The findings of this study have direct implications for language educators and curriculum designers who aim to support students' academic success and emotional well-being. First, the significant role of FLLE in fostering intrinsic motivation and reducing burnout points to the need for emotionally supportive learning environments. Teachers should actively create opportunities for enjoyment in language learning by integrating dynamic, engaging activities that enhance both social and private aspects of enjoyment. Examples include peer collaboration, interactive discussions, multimedia resources, and culturally rich content, all of which have been shown to foster positive emotional engagement. Additionally, fostering positive teacher-student relationships, providing constructive feedback, and encouraging participation in less formal language learning activities (e.g., language clubs) can amplify the positive effects of FLLE, creating a more supportive and motivating classroom climate.

Second, the mediating role of L2 grit highlights the need to go beyond content-based language teaching by focusing on developing students' resilience and perseverance. Educators should encourage long-term goal setting, both personal and academic, that fosters persistence through challenges. Rather than emphasizing immediate academic outcomes, such as test scores, teachers can help students frame their language learning in the context of lifelong goals. This could include incorporating reflective activities where students set personal language learning goals, track their progress, and learn from setbacks. Furthermore, educators can instill a growth mindset by encouraging students to view challenges as opportunities for growth, which is critical for cultivating grit.

Finally, the clear link between academic burnout and reduced motivation underscores the importance of prioritizing student well-being in language programs. Schools and universities should implement resources such as counseling services, peer support networks, and stress management workshops to address the emotional and psychological challenges that can arise in high-stakes academic environments. Offering more flexible learning pathways, integrating wellness programs, and creating a more balanced academic workload are practical steps institutions can take to alleviate the pressures that contribute to burnout. Additionally, instructors can play a proactive role by recognizing signs of burnout early and providing individualized support, ensuring students remain motivated and engaged.

Limitations and directions for further research

Although this study provides meaningful insights into the relationships among FLLE, academic burnout, L2 grit, and motivated behavior, several methodological limitations should be acknowledged. First, the quantitative

phase utilized convenience sampling, which, although practical, may introduce biases that affect the generalizability of the findings. To mitigate this limitation, participants were recruited from multiple universities and major cities to ensure diversity in socioeconomic backgrounds, proficiency levels, and learning experiences. However, future research could improve representativeness by employing probability sampling methods.

Second, the qualitative phase used purposive sampling to capture a range of learner experiences. While effective for exploring diverse perspectives, the relatively small sample size constrains the generalizability of these findings. Expanding the qualitative sample or adopting longitudinal designs could offer a deeper understanding of how learners' emotional and motivational dynamics evolve over time. Third, the cultural and educational focus on EFL learners in mainland China may limit the applicability of the findings to other contexts. Cultural and institutional factors significantly shape language learning experiences, and these variables may differ across regions. Future research could address this limitation through cross-cultural studies to identify both universal and context-specific patterns in emotional and motivational processes. Finally, while this study concentrated on FLLE, grit, burnout, and motivation, additional psychological constructs such as self-regulation, emotional intelligence, and self-efficacy warrant further investigation. Exploring these factors could provide a more holistic understanding of resilience and sustained motivation in language learning contexts.

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Author contributions

Yao Song is the sole author of this manuscript and contributed to the design, data collection, analysis, and writing of the paper.

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Data availability

The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with the ethical guidelines and principles of the Declaration of Helsinki. The research protocol was approved by the Institutional Review Board of Henan University of Animal Husbandry and Economy. All participants provided informed consent prior to participation in both the quantitative and qualitative phases of the study.

Consent for publication

Not applicable.

Competing interests

The author declares no competing interests.

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